Kathmandu University

School of Engineering

Department of Artificial Intelligence

Final Year Project Presentation

MarketGyan

NEPSE and Financial News Analysis

An Extension of Khabar AI

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Outline

Introduction

Project idea

MarketGyan extends Khabar AI into a Nepal-focused financial NLP assistant for NEPSE-related news, evidence search, and grounded market reports.

- Connects market snapshots, financial news, and policy documents.
- Classifies whether a story is directly relevant, indirectly relevant, or not relevant to NEPSE.
- Produces structured impact fields: event type, direction, sector, symbol, mechanism, and evidence sentence IDs.
- Uses RAG so current facts remain in retrieved evidence rather than model weights.

Motivation and Background

Why this matters

- NEPSE data, financial news, and regulatory updates are scattered across many sources.
- Investors must manually connect market movement with related events.
- Generic news summarizers do not know NEPSE-specific symbols, sectors, and event taxonomy.

Need for the system

A Nepal-specific assistant should answer with source-backed evidence, not generic positive or negative tone.

Safety boundary

Informational analysis only. No buy/sell advice.

Project Objectives

Main objective

Build and evaluate a Nepal-specific financial news analysis system that combines structured NEPSE data, financial-news retrieval, and a fine-tuned instruction model.

Specific objectives

1. Collect and manage NEPSE-related market data, financial news, and selected regulatory sources.
2. Create a bilingual, evidence-grounded NEPSE impact benchmark.
3. Fine-tune and evaluate Qwen3.5-9B LoRA for compact structured impact extraction.
4. Build a RAG runtime for grounded query answering and report generation.
5. Evaluate schema validity, grounding, F1 scores, retrieval precision, latency, and citation correctness.

Scope of the Project

Included

- Public NEPSE snapshots and finance news.
- English and Nepali article processing.
- NEPSE-Impact-500 dataset.
- Qwen3.5-9B LoRA adapter.
- Evidence search, Ask MarketGyan, and report generation.

Not included

- Brokerage integration or order execution.
- Personalized investment advice.
- Trading prediction or causal market forecasting.
- Production deployment until retrieval and content gates pass.

Literature Review: Financial NLP

FinBERT

FinBERT shows that adapting transformer models to financial text improves financial sentiment classification compared with generic language models.

- Useful lesson: financial words are event- and domain-dependent.
- MarketGyan gap: NEPSE analysis needs more than positive/negative sentiment.
- Required fields include relevance, event type, impact scope, direction, sector, symbol, and evidence.

Literature Review: Financial LLMs

Study	Relevance to MarketGyan
FinGPT	Supports a data-centric approach: curate financial data and adapt a moderate-size model with lightweight tuning.
FinLLM survey	Highlights hallucination risk, benchmark weakness, and the need to separate validity, grounding, and task metrics.
FinTral	Demonstrates financial instruction tuning for multi-task structured outputs on a moderately sized open model.

Adopted design

MarketGyan treats the Nepal-specific ontology and evidence-grounded benchmark as the main contribution; LoRA is the adaptation method, not the novelty.

Literature Review: Nepali NLP

Work	Relevance and limitation
NepBERTa	Shows that Nepali-specific pre-training improves Nepali NLP tasks.
Subedi et al.	Evaluates LLMs for Nepali NER and POS tagging, showing promise but also low-resource benchmark gaps.
Thapa et al.	Expands Nepali transformer resources with larger monolingual pre-training for understanding and generation.

Gap

These works are not finance-specific. MarketGyan therefore builds a bilingual NEPSE benchmark instead of relying only on general Nepali NLP resources.

Literature Review: Efficient Adaptation

LoRA

- Trains low-rank adapter matrices instead of all model weights.
- Makes domain adaptation practical for a project-scale GPU budget.
- Used for the final Qwen3.5-9B adapter.

QLoRA

- Reduces memory through quantized base-model loading.
- Useful for early diagnostic Qwen3-8B runs.
- Not used as the official final candidate.

Proposed System Overview

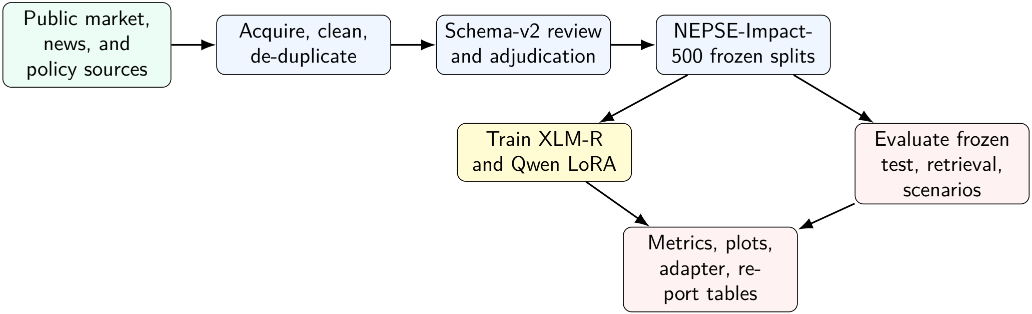
Core pipeline

1. Collect market/news/policy data.
2. Normalize and index finance evidence.
3. Retrieve candidate evidence for a query or report date.
4. Ask Qwen3.5-9B for compact structured output.
5. Materialize citations from retrieved rows.

Final runtime decision

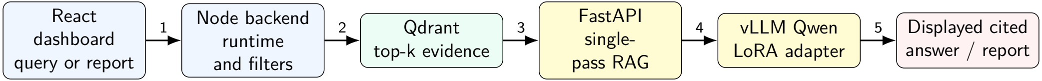
Single-pass RAG replaced the earlier context-heavy multi-agent prompt because it passed live citation checks without exceeding the vLLM context limit.

Training and Evaluation Workflow



The frozen benchmark keeps model comparison separate from live retrieval and frontend behavior.

Daily Inference Workflow



Citations are reconstructed from retrieved evidence rows with URL, excerpt, content hash, chunk ID, and sentence IDs.

Methodology I: Data and Schema-v2

Data pipeline

- Acquire public NEPSE snapshots, financial news, and selected machine-readable policy text.
- Clean, de-duplicate, sentence-split, and attach stable URLs/content hashes.
- Review generated candidates and export only adjudicated gold labels.

Benchmark

- NEPSE-Impact-500: 300 direct, 100 indirect, 100 hard negatives.
- Frozen split: 350 train, 75 validation, 75 test.
- Duplicate groups stay inside the same split to prevent leakage.

Methodology II: Schema-v2 JSON Contract

```
{
  "relevance": "direct | indirect | not_relevant",
  "eventType": "market_trading | earnings | regulation | governance |
                credit_financing | capital_action | sector_industry |
                project_operations | macro_policy | other | none",
  "impactScope": "market | sector | company | not_applicable",
  "impactDirection": "bullish | bearish | neutral | uncertain |
                      not_applicable",
  "sectors": ["Banking"],
  "symbols": ["NABIL"],
  "rationale": "short evidence-based reason",
  "confidenceBand": "low | medium | high",
  "evidenceSentenceIds": ["S1", "S3"]
}
```

Validation rule

Evidence IDs must point to real numbered source sentences. Relevant rows need defensible impact fields; not-relevant rows must not invent sectors, symbols, or market impact.

Methodology III: Modeling and Runtime Evaluation

Modeling

- XLM-R relevance and direction baselines.
- English-only FinBERT sanity baseline.
- Base Qwen zero-shot and three-shot ablations.
- Qwen3.5-9B BF16 LoRA targeted-v2 final adapter.

Evaluation

- Frozen test split for model metrics.
- vLLM JSON Schema constrained inference check.
- Retrieval Precision@5 over 30 proposal queries.
- Live scenario and report-generation smoke tests.

System Requirements and Training Setup

Area	Final setup from report
Local runtime	React, Node/Express, MongoDB, Qdrant, Python FastAPI agent.
Model serving	Lightning AI Studio with vLLM OpenAI-compatible endpoint.
XLM-R baseline	`xlm-roberta-base`, 512 tokens, LR $2e{-5}$, max 8 epochs, class-weighted loss, validation macro-F1 selection.
Qwen training	`Qwen/Qwen3.5-9B`, BF16 LoRA, rank 32, alpha 64, max sequence 2048, LR $2e{-5}$, 3 epochs, A100 80GB.
Safety	Local report generation, citation validation, disclaimer, no buy/sell advice.

Qwen3.5-9B Training Configuration

Item	Value	Item	Value
Base model	`Qwen/Qwen3.5-9B`	Precision	BF16 LoRA, no 4-bit
Max sequence	2048	Max generation	256 tokens
LoRA	rank 32, alpha 64, dropout 0.05	Epochs	3
Batch	train/eval 4	Grad accumulation	2
Learning rate	$2e{-5}$	Optimizer	`paged_adamw_8bit`
Checkpointing	eval/save every 10 steps	Best model	validation loss
Oversampling	`targeted_v2`	Train rows	350 to 922 weighted examples
Output	`marketgyan-qwen35-9b-a100-80gb-bf16-lora-targeted-v2`		

Reason for this setup

The latest run optimized content metrics while preserving strict JSON validity and evidence grounding; the frozen test split was not used for oversampling.

Tools and Technologies

Layer	Tools used
Frontend	React dashboard with MarketGyan tabs: Overview, Ask, Evidence Search, Reports, System.
Backend	Node.js / Express APIs, MongoDB, scheduler, report persistence, review tools.
Retrieval	Qdrant vector database with finance-specific evidence payloads.
ML service	Python, FastAPI, evaluation CLI, schema validators.
Model training	Qwen3.5-9B, LoRA, BF16 training, A100 80GB, frozen test split.
Model serving	Lightning AI Studio, vLLM OpenAI-compatible endpoint, JSON Schema constrained decoding.

Implementation Details

Backend and data modules

- Article, market snapshot, market document, security, and report records.
- Candidate generation, review, adjudication, and schema validation.
- Runtime status, protected report generation, scheduler, and persistence.
- React tabs for Overview, Ask, Evidence Search, Reports, System, and validation.

ML and runtime modules

- XLM-R baseline training and Qwen LoRA notebook/script pipeline.
- Qdrant finance-evidence indexer with source URL, content hash, chunk ID, and sentence IDs.
- FastAPI single-pass RAG agent for grounded answers and reports.
- Evaluation CLI for frozen model metrics, retrieval labels, and live scenarios.

Results and Evaluation

Metric	Target / Gate	Observed	Status
Structured validity	≥ 0.95	1.000	Pass
Evidence grounding	≥ 0.95	1.000	Pass
Relevance macro-F1	≥ 0.78	0.796	Pass
Event macro-F1	≥ 0.65	0.633	Below target
Direction macro-F1	≥ 0.65	0.575	Below target
Sector F1	≥ 0.76	0.677	Below target
Symbol F1	≥ 0.81	0.793	Below target
Evidence sentence F1	≥ 0.77	0.736	Below target
Live scenarios	20/20	20/20	Pass
Live latency	Reported	avg 6.665s, p95 9.735s	Demo-ready
Retrieval Precision@5	≥ 0.60	0.262 judged, 0.113 all-query	Fail

Baseline and Qwen Ablation Results

Supervised baselines

Model / task	Acc.	Macro-F1
XLM-R relevance	0.787	0.723
XLM-R direction	0.683	0.549
FinBERT direction	0.889	0.616

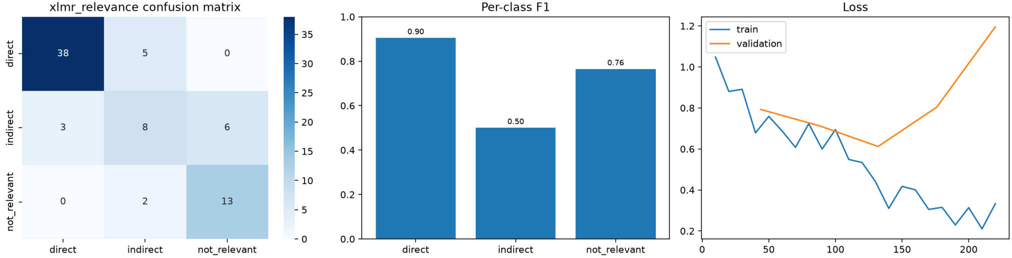
FinBERT support is small and English-only; XLM-R is the main multilingual classifier baseline.

Qwen conditions

Condition	Valid	Rel. F1	Evid. F1
Zero-shot	0.000	0.367	0.000
Three-shot	0.987	0.435	0.599
Targeted-v2 LoRA	1.000	0.796	0.736

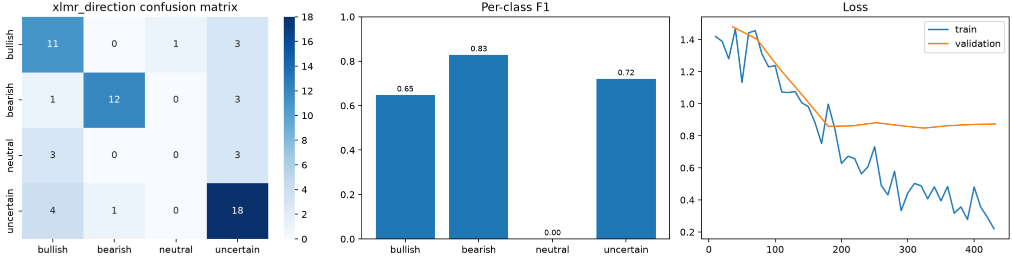
Fine-tuning solves strict structure/grounding and improves relevance, but neutral direction remains weak.

XLM-R Relevance Baseline Plot



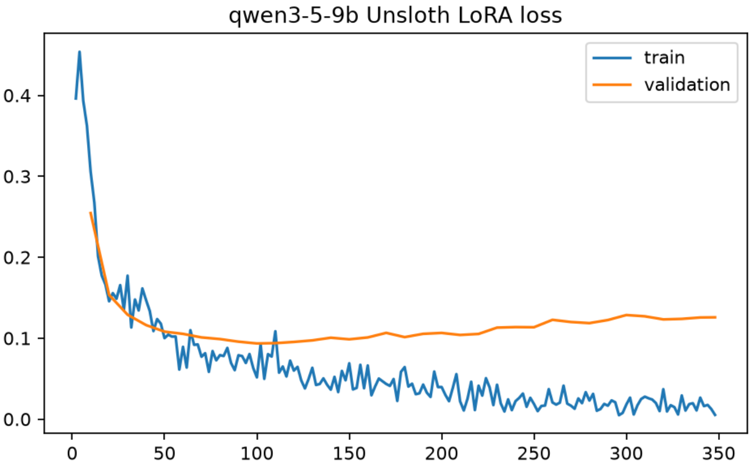
XLM-R relevance classifier: accuracy 0.787 and macro-F1 0.723 on the frozen test split.

XLM-R Direction Baseline Plot



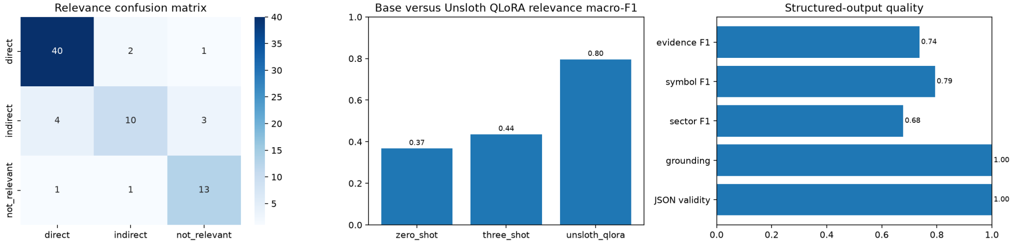
XLM-R direction classifier: accuracy 0.683 and macro-F1 0.549 on relevant frozen-test rows.

Qwen Targeted-v2 Loss Curve



Validation loss flattens and rises after early improvement, supporting best-checkpoint selection instead of longer blind training.

Qwen Targeted-v2 Test Result Plot



Frozen-test diagnostic plot from the final Qwen3.5-9B targeted-v2 adapter artifact.

Constrained Inference and Retrieval Readout

Area	Primary result	Comparison	Interpretation
Constrained validity	0.987	adapter 1.000	Still above schema gate.
Constrained grounding	0.987	adapter 1.000	Still above grounding gate.
Constrained relevance F1	0.819	adapter 0.796	Slight relevance gain.
Retrieval judged P@5	0.262	target 0.600	Proposal retrieval gate fails.
Retrieval all-query P@5	0.113	target 0.600	Empty filtered queries expose metadata/indexing gap.

Decision impact

JSON Schema constrained decoding is useful for model-only structure, but retrieval metadata must improve before deployment.

Demo and Key Features

Ask MarketGyan

- Grounded answer from single-pass RAG.
- Disclaimer and no-advice boundary.
- Citations with title, URL, excerpt, chunk ID, and sentence IDs.

Evidence Search

- Query finance evidence directly.
- Inspect source, sector, symbol, language, and date filters.
- Verify expanded sentence evidence.

Reports

- Generate local report for latest snapshot.
- Published 2026-06-11 report.
- 3 citations and 3 sector-analysis rows.

Limitations and Future Work

Limitations

- Retrieval Precision@5 fails the proposal gate.
- Neutral direction recall is 0.000.
- Sector, symbol, and evidence recall need improvement.
- Historical snapshots are partial.
- Second-review agreement is not yet reported.

Future work

- Normalize sector/source/ticker metadata in Qdrant payloads.
- Rerun 30-query retrieval evaluation.
- Add targeted neutral and ticker hard cases.
- Complete second annotation and agreement metrics.
- Keep single-pass RAG as the safer serving boundary.

Conclusion

Final contribution

MarketGyan delivers a Nepal-specific, evidence-grounded financial NLP research prototype with NEPSE-Impact-500, Qwen3.5-9B targeted-v2 LoRA, Qdrant evidence retrieval, and a live React/FastAPI/Node demonstration path.

- The model passes strict JSON validity and evidence grounding.
- The live single-pass RAG system passes citation, disclaimer, and no-advice scenario checks.
- The frontend can demonstrate Ask, Evidence Search, and Reports.
- Deployment is not recommended until retrieval and content metrics improve.

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